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Frontend Q/A

1. Javascript and Features

*JS is a progeramming language used to create web applications

* It is opensource and dyramically typed

*JS is single thread architecture

* JS is used to manipulate the datas dynamically and it uses Dom to update HTML elements

Features.

* Single Thread Architecture stops execution even it has single error

* API Access

* Dynamically Typed

* Interpreted Language.

2. Functions.

* It is a block of code we can use it again (reusable) that can be called after declaration.

* bodes needs to be written inside curly braces { }.

* Function we can declare data inside function or can be passed as an argument

* Argument can be a value, variable or another function.

* Return statement should be there inside a function

Type:

function declaration

1. function Count () {

let a = 1

count = a+1

return count

}

2. Expression -> declared as variable

let count = function () {

}

3. Arrow function

const consist = () => {

return

}

Spread operator.

* Spread operator : "..."

* It used to make an duplicate array because mostly we can't make changes in react

* It is used inside function when we the no. of data is unknown

* using this we can merge arrays into array

* let duplicate = function (...arr) { }

* Two arr = [...arr1, ...arr2]

4. React

* React is a JS Library used to build web applications.

* It use component based architecture

* React was developed by meta in 2013

* we have React 3 and Vite 2 versions.

* previous versions use ExLint current version use OxLint.

* It uses V-Dom insted of Dom

* Virtual Dom (V-Dom) is lite version of Dom

* Using V-Dom we can update Re-render a specific component not whole webpage.

5) JSX

* JSX - Javascript XML [External Markup]

* It use HTML inside javascript using fragments inside return

* It still reads only JS file (the compiler) this file will be translated by OXLint (ExLint perviously)

* Each file contains a main function as main export

6. Events:

* Event is a process that happens after interaction is made to like typing clicking and all

* On click, On change, On Submit, OnKeyDown, OnBlur, OnKeyUp and OnFocus

* Declared at end of opening tag

* Events triggered using a function;

components:

* components are small part of react webpage

* It is reusable code used to design webpage

* It is created in jsx file

* Components must start with capital letter when exported <Component/> and used as tag

* It jsx file should contain one default function and normal functions.

Export

* Default Export - only one

const function App() {

}

export default App

* normal Export - Multiple in a file

export const App() {

}

Import

*used to call the exported components

Default Export file

import Name from "path";

Normal Export

import {Name} from "path";

8. Rendering.

* Rendering is the process do the updates in webpage without loading

* once a component is being update, the update in the UI is visible only when its corresponding component got re-render.

Types

Primitive Rendering

Rendering variables like Integer, String Boolean values

* String and Integers can be stored in a variable and simply called in {} will render it in web page

```
let string = "Name"

My {string}     is DS
let nums = 4
The number is {nums}
```

*Boolean

* Optional

```
let val = null
{val && "hello"}
```

*Nullish collesion

```
"React" => val = 'null';
"bthu   => val = "otem
```

Non-Primitive

* Renders variable like array, object and array of object

```
arr
{arr.map ((e,i) => (
    <p>{e}</p>
))}
```

object

```
<div>
    <p>{obj.name}</p>
    <p>{obj.age}</p>
```

arr of obj.

```
{map((e,i) =>(
    <li key={i}>
        <p>{e.name}</p>
    </li>)
```

9. Hooks

* It is the special function inside functional components

* It processes update and render components.

* It handles API calls, Local storage and dynamic operations

* Hooks is useState.

10. State

* This data management inside the component.

* It is similar to variable

* State will re-render the component with out refreshing pages.

* State accessed using useState

Example:

```
const [state, useState] = useState("iv")

    const status = () => {
        SetState("uv")
    }
    return (<>
        <p>{state}</p>
        <button onClick={status}>
```

props:

* It is passes data from one component to another as object.

* props are read only temporary memory

```
const function = () => {
    return <    />
}
const func = ({name}) => {
    <p>{name.name}</p>
    <p>{name.age}</p>
}
```

* parent component contains data calling the child component
* Child component cannot update the data.